

RESEARCH ARTICLE

Non-verbal behaviour of professional soccer players performing in the absence or presence of fans

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Abstract

The present research tested the impact of score line, spectator presence and game location on athletes' non-verbal behaviour (NVB). Two studies ($N = 240$; $N = 160$) tested whether the NVB of soccer players differed as a function of score line, spectator-presence and game location by asking participants to guess the current score, the dominance, pride and confidence of soccer players in videos. Results demonstrated that NVB ratings were more negative when the athletes were trailing (less dominant, less proud, and less confident) compared to leading. Study 1 showed more non-verbal expressiveness (social facilitation) of pleasant feelings (being in a high lead) and lower expressiveness (social inhibition) of negative feelings (trailing) in the presence of spectators. Study 2 indicated that leading at away games led to higher non-verbal expressiveness, whereas leading at home games led to less non-verbal expressiveness in the presence of a supportive audience.

KEYWORDS

body language, emotion expression, social facilitation, social inhibition, sport, thin slices

1 | INTRODUCTION

One of the oldest questions in psychology is how the presence of other people affects an individual's or a group's behaviour (Steinmetz & Pfattheicher, 2017; Triplett, 1898; Zajonc, 1965). Within the context of sports, there is no doubt that the presence of cheering or booing spectators in the large arenas especially built for these events can have an impact on athletes (e.g., Anderson et al., 2012; Fothergill et al., 2014). The Covid-19 pandemic has led to unique conditions to investigate the effects of the absence of spectators in sports competitions in a natural experiment. The present research has made use of this opportunity to investigate the influence of sport spectators on the non-verbal behaviour (NVB) of professional soccer players. Prior research on NVB in sports has shown that the NVB of athletes is influenced by contextual factors like the current score line (Furley & Schweizer, 2014) and that observers can make accurate

inferences about the score based on thin slices of the athletes' NVB. In the present series of studies, we extend this earlier research by comparing observers' assessments of athletes' NVB during normal and 'ghost matches' and thereby contribute to an important question in psychology: how the attention of others affects people's behaviour.

According to athletes, coaches, referees and fans themselves, sport spectators are a crucial factor contributing to success (e.g., winning a match or performing well) in sporting events (Anderson et al., 2012; Fothergill et al., 2014). While it is certain that the presence of spectators has an influence on athletes, the effects are not always positive. Triplett (1898) made one of the first notable observations in this context when he realized that cyclists performed better when competing against another person as opposed to cycling alone. This study laid the foundation for research on social facilitation, a term coined by Allport in 1924. While social facilitation first referred to performance

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increases in co-acting situations, it was soon confirmed that the effects even occur in the mere presence of passive observers. The mere presence of others was found to influence performances in various types of tasks, for example, jogging (Strube et al., 1981), basketball free throws (Kotzer, 2007) or playing pool (Michaels et al., 1982). However, studies have found not only facilitation but also social inhibition effects in the presence of others. A meta-analysis (Bond & Titus, 1983) and reviews (Paul, Adele & Andrew, 2022; Wann & Hackathorn, 2019) of audience effects on performance show that whether the influence of spectators on athletes' performances is positive or negative seems to depend on the task (Strauß, 2002) and also on the type of sports being performed. In soccer, the influence of spectators on players' performance as well as on referees' decision making are seen as factors in explaining the home advantage (Jamieson, 2010; Pollard & Gómez, 2014; Staufenbiel et al., 2015). While a supportive audience can foster the motivation of the home team, it can also cause self-monitoring and overcautious actions leading to choking under pressure (Wallace et al., 2005). Lately, Calleja et al. (2022) have shown that soccer players reported feeling pressured and having anxious thoughts as a result of spectator behaviour, contradicting the home advantage. Pertinent to the present research, studies have shown that the aforementioned states of mind, emotions and contextual factors can be correlated with the players' NVB, and are visible to observers (e.g., Furley et al., 2018) and thereby contribute to social facilitation or inhibition on performance.

Consequently, the present study addressed the NVB of athletes as a function of the presence or absence of spectators. The *American Psychological Association* (n.d.) defines NVB as actions that provide information about an individual's internal state without speech. NVB is considered a subprocess of non-verbal communication (Hess, 2016; Matsumoto, Frank & Hwang, 2013). It encompasses a wide range of behaviours as all movements can be considered expressive to some degree including facial, vocal and postural expressions, as well as touch, proxemics and gaze (Wiener, Devoe, Rubinow & Geller, 1972). Several theoretical accounts predict that humans are highly efficient at communicating internal states (Bogdan, 1997) like emotions (Darwin, 1872; Ekman, 1992) or social intentions (Fridlund, 1994) non-verbally. Pertinent to the present research, the Behavioural Ecology Theory of NVB (Crivelli & Fridlund, 2018) was developed to account for audience effects on a person's NVB. A main tenet of the theory is that a person's NVB is shaped by a range of personal and contextual factors, and non-verbal displays do not necessarily have to be functional to the individual showing the display. That is, non-verbal displays may be intentional or unintentional and are believed to guide further social interaction with observers—for the better or the worse (Crivelli & Fridlund, 2018; Fridlund, 1994).

Of particular importance to the present research, previous studies in the domain of NVB have focused on emotional expressions when people are alone compared to when people are in the presence of others (Philippot, Feldman & Coats, 1999 for a review). These studies have provided mixed results. On the one hand, studies have suggested that social presence increases expressiveness while other

studies have shown that the presence of observers leads to decreased expressiveness (see Wagner & Lee, 1999 for an overview). For example, an early study by Andrus (1946) showed that the frequency of laughter increased linearly with increasing audience size. Similarly, students were shown to laugh and smile more at jokes when listening to the jokes together with other students than when they were alone (Young & Frye, 1966). The same results emerged in a series of studies (Chapman, 1973, 1975; Chapman & Wright, 1976) with 7–8-year-old children who smiled and laughed more when other children were watching. Several further studies have provided evidence for similar social facilitation in a variety of different contexts (Chovil, 1991; Hess, Banse & Kappas, 1995; Jakobs et al. 1999a, 1999b). In an important series of studies, using an ethological approach to assess smiling behaviour in naturalistic contexts, Kraut and Johnston (1979) showed that people in general and athletes (bowlers) in particular were non-verbally more expressive (i.e., showed more smiles) when socially engaged than in the absence of other people. Additionally, a seminal study by Fridlund (1991) demonstrated that non-verbal facial behaviour was better predicted by the social context than by experienced emotions as participants showed more smiles in response to pleasant video tapes when other people were present, watching, or they merely believed that others were watching compared to when they were alone.

On the other hand, Kraut (1982) exposed participants to a variety of pleasant and unpleasant odours either alone or in the presence of others. When other people were present, non-verbal facial behaviour was less spontaneously revealing of odour evaluations. A further series of studies showed that children's NVB was less expressive when other people were present (Kilbride & Yarczower, 1980; 1983; Yarczower, Kilbride & Hill, 1979). Yarczower & Daruns (1982) further showed that the presence of a potential communication partner led to the suppression of facial expressions in children. In addition, Friedman & Miller-Herringer (1991) provided evidence that people expressed less pleasure in response to winning a competition in the presence of the defeated confederates than in a non-social situation. Several further studies have provided evidence for similar social inhibition in a variety of different contexts (Costa, Dinsbach, Manstead & Bitti, 2001; Guerin, 1989; Kleck et al., 1976).

Buck, Losow, Murphy & Costanzo (1992) suggest that a distinguishing feature between studies showing social facilitation or social inhibiting of non-verbal expressions might be the affective nature of the situations with pleasant feelings coinciding with social facilitation and negative feelings with social inhibiting. However, this is most likely not the only factor contributing to social facilitation or inhibiting, as other factors like the role or relationship of the interacting or observing people showed to affect non-verbal expressiveness (Wagner & Lee, 1999 for an overview).

Hence, the reviewed research suggests that the effects of observers on other people's (non-verbal) behaviour is complicated and is influenced by a range of internal (e.g., valence of emotion or feeling) and external (e.g., relationship with observer) variables. Therefore, previous research does not seem to allow straightforward predictions of how the presence or absence of spectators in

professional sports would generally affect the non-verbal behaviour of athletes.

2 | THE PRESENT RESEARCH

Within the present research, we adopted a similar thin-slices paradigm (Ambady & Rosenthal, 1993; Ambady et al., 2000) as earlier research in sports did (Furley & Schweizer, 2016). The thin-slices approach has accumulated evidence that people can draw a range of accurate inferences based on short observations of other peoples' NVB (Ambady et al., 2000; Carney et al., 2007). Research in the context of sports has further shown that different NVBs are displayed in response to successful or unsuccessful performances in sports (e.g., Aviezer et al., 2012; Tracy & Matsumoto, 2008) and that observers can accurately interpret this NVB (Furley & Schweizer, 2014, 2016). These findings were understood as evidence that trailing during sport competitions (i.e., having a lower score than the opponent) led to negative feelings (e.g., low confidence and feeling ashamed) and leading to positive feelings (e.g., boosted confidence and feeling proud), which was assessed by observers as lower/higher status-related NVB respectively (Furley & Schweizer, 2016). The first study aimed at examining the influence of spectator presence on observer assessments of the score related NVB of soccer players.

We adopted the approach of Furley and Schweizer (2014, 2016) and asked participants to guess the current score and rate the dominance, pride and confidence of a soccer player in a short video clip taken from a soccer match (without knowing the actual score in the video clip). The goal of Study 1 was to use this evaluative coding technique as an indirect measure of an athlete's NVB (Bente, Senokozlieva, Pennig, Al-Issa & Fischer, 2008) to test how the presence/absence of spectators affects NVB during sport competitions. The assumption was that athletes' NVB would differ depending on whether spectators were present or absent during the match. Consequently, these differences in athletes' NVB should be reflected in participants' differing guesses of the score and evaluations of dominance, pride and confidence as a function of spectators' presence.

The first main hypothesis (1) we tested was replicating the findings of Furley and Schweizer (2014; 2016). It was hypothesized that participants are able to accurately guess the score and would rate the NVB as more dominant, proud and confident when the athletes were leading as opposed to trailing (without knowing the actual score in the video clips). The second research question regarding the influence of spectators on score guesses and NVB ratings was more explorative. Nevertheless, our second hypothesis (2) was that participants would be able to accurately guess the score and would rate the NVB as more dominant, proud and confident when the athletes were leading as opposed to trailing regardless of whether clips were taken from normal or ghost matches (i.e., regardless of whether spectators were present or not during the match). This hypothesis was based on the evolutionary reasoning of winning and losing NVB in sports (Furley, 2019; Furley & Schweizer, 2016). According to this perspective changes in athletes' NVB when winning or losing in sports resemble the adaptive

signalling of primates when winning or losing status contests (Furley, 2019; Furley & Schweizer, 2016): losing a contest coincides with submissive non-verbal expressions (i.e., low social status) and winning with dominant and proud NVBs (high social status). If this evolutionary reasoning is correct, then we would arguably expect to find similar effects of the actual score line during the clips on participants' ratings of athletes' NVB. However, in light of the outlined research on social facilitation and inhibition of emotional expressions, it also seems feasible that the presence of spectators will affect athletes' NVB. In this respect, it seems reasonable to assume that leading athletes are experiencing more positive feelings and trailing athletes more negative feelings. Hence, we tried to further test the reasoning of Buck et al. (1992), who suggested that pleasant feelings coincide with social facilitation (i.e., greater expressiveness of positive NVB) and negative feelings with social inhibiting (i.e., less expressiveness of negative NVB) in response to the presence of observers. Therefore, we hypothesized (hypothesis 3) that the presence of fans in the stadium would lead to higher evaluations (social facilitation) of athletes' status-related NVB (higher score guess, higher dominance, pride and confidence rating) in the high leading category compared to when playing in empty stadiums. On the other hand, we expected lower evaluations (social inhibition) of athletes' status-related NVB (lower score guess, lower dominance, pride and confidence rating) with fans present in the high trailing category compared to when playing in empty stadiums.

The second study aimed to extend the first study as previous research has shown that the game location (home vs. away) further influences the assessment of NVB of athletes (Furley et al., 2018). Therefore, Study 2 added the additional independent variable of game location (home vs. away) to the independent variables of Study 1 (presence of spectators and current score line). In this respect, we further explored differential effects of a supportive crowd (home games) compared to a non-supportive crowd (away games) as previous research has suggested that the relationship a person has with an observer can affect the expression of certain NVBs (Wagner & Lee, 1999 for an overview). Hence, one might speculate (hypothesis 4) that a supportive crowd might lead to a higher expressiveness of positive feelings when leading and negative feelings when trailing as research (summarized in Wagner & Lee, 1999) has shown that people are more comfortable expressing their feelings among supportive friends compared to strangers. However, as this is highly speculative and it remains unclear if this theorizing transfers to the context of home and away games in soccer, we investigated the effect of home and away games in an explorative manner by exploring potential main effects and interactions with the other independent variables score line (trailing = negative feelings, draw = neutral feeling, leading = positive feelings) and amount of social attention/feedback the athletes were receiving (fans = a lot of attention/feedback, no fans = little attention/feedback).¹

¹ We are aware that it is a simplistic assumption that trailing always coincides with negative feelings, drawing with neutral feelings and leading with positive feelings. However, we consider it a reasonable simplification that on average these three score categories should be differentially associated with negative, neutral and positive feelings.



FIGURE 1 Sample frame of the video stimuli used in the present research.

3 | STUDY 1: SCORE ESTIMATION AND EVALUATIVE CODING OF NVB IN SOCCER

Sampling representative stimulus material was crucial for the current investigation and has been described as challenging (Fiedler, 2011), especially using the thin-slices paradigm (see Ambady et al., 2000 for an overview). Therefore, we initially ran a series of pilot studies before running the reported studies. Full details of the pilot studies can be found in the online appendix.

3.1 | Method

3.1.1 | Stimulus material

For the purpose of this study we sampled stimulus material from ten matchdays of the 2019/2020 season of the highest German soccer league (i.e., the Bundesliga). We deliberately choose this season as it constitutes somewhat of a natural experiment as all matches from matchday 26 were postponed and played without spectators due to the Covid-19 pandemic. Footage was available from five matchdays before the obligatory 'Corona break' (matchdays 21–25) and five matchdays after the break (matchdays 26–30) and hence from a total of 90 matches. However, spectators were present at only 43 of those 90 matches due to Covid-19 regulations. Thus, video material was collected from a total of 86 matches; that is, 43 normal matches with spectators and 43 ghost matches without spectators.

All stimuli meeting the following criteria were selected from the available video footage. All scenes throughout the entire playing time were considered. Only close-ups of players (i.e., no coaches, referees, etc.) were collected. Each clip clearly focuses on one player (see Figure 1 for an example); if several players were in focus simultaneously or the focus shifted between players, the clip was excluded. In the fashion of Furley and Schweizer (2014) only expressions of NVB during game interruptions (e.g., throw-ins, free kicks or corner

kicks) were chosen in order to ensure that the results would not be influenced by score-related changes in sport-specific behaviour, such as strategy and tactics. Clips showing players in the context of substitutions, injury/treatment breaks, video replays, penalties or scored goals were excluded. Scenes of players who just got a yellow/red card were also excluded. This was done to avoid confounding effects on participants' score ratings by excluding scenes which show players' direct (emotional) reactions to the respective game events (e.g., anger about refereeing decisions). Therefore, any obvious non-verbal signals associated with anger, disappointment, frustration, or the like, such as players shaking their heads, yelling or punching the air (cf. Witkower & Tracy, 2019), were carefully avoided. Additionally, obvious non-verbal expressions of pride and shame, such as players raising their fists above their heads or covering their faces with their hands (cf. Tracy & Matsumoto, 2008), were avoided as they would be too obvious signs associated with winning and losing (Furley & Schweizer, 2014). Finally, none of the clips gave any indication of whether spectators were present during the match or not (i.e., whether the clip was taken from a normal or a ghost match).

The selection process of the stimuli was aided by two research assistants, blind to the purpose of the study. To ensure an unbiased selection procedure, the selection criteria for the video clips were defined as precisely as possible beforehand. An interrater agreement of at least 80% was determined as prerequisite for the continuation of the material selection (as commonly recommended, cf. McHugh, 2012). A test run was performed on one of the matches to ensure agreement between the two research assistants on the selected stimuli. After an initial agreement of 70% of the selected scenes, the selection criteria were revised. Another test run was performed on a different match, leading to an agreement of 85% of the selected video clips.

According to these criteria, a total of 3464 clips with a mean duration of 3.2 seconds ($SD = 1.7$ s) were collected. Of this entire pool, 1513 clips were taken from normal matches ($M = 3.1$ s; $SD = 1.7$ s) and 1951 clips were taken from ghost matches ($M = 3.3$ s; $SD = 1.8$ s).

3.1.2 | Experimental categories

The within-subject manipulation of the present study was based on the different categories of video clips. Ten categories of stimuli were distinguished in the experiment. First, a distinction was made between normal matches with fans present in the stadium and ghost matches where no fans were allowed. Furthermore, five different score categories were defined: (1) high lead, referring to clips showing players whose team is leading by at least two goals (161 clips with spectators; 154 clips without spectators); (2) close lead, showing players in a close game situation whose team is leading by only one goal (230 clips with spectators; 283 clips without spectators); (3) tied score/draw, showing clips of players while the score is tied (714 clips with spectators; 1050 clips without spectators); (4) close behind, showing players in a close game situation whose team is trailing by only one goal (232 clips with spectators; 306 clips without spectators); (5) high behind, referring to

clips showing players whose team is trailing by at least two goals (176 clips with spectators; 158 clips without spectators).²

3.1.3 | Participants

A total of 241 participants (128 male, 110 female; $M_{age} = 29.42$ years; $SD = 17.46$ years; there was missing data on some of the demographics for 3 participants) participated in Study 1. Participants gave written informed consent before the experiment. 117 participants had no active soccer experience, 87 reported to have some recreational soccer experience, and 37 played on a team (2 of them professionally). The participants consumed an average of 25 h ($SD = 49.2$) of soccer per year on television. Participants were recruited on a voluntary basis by research assistants.

As the present approach of assigning every participant a unique random set of stimuli would likely lead to more between-participant variations, we substantially increased the sample size compared to Furley and Schweizer (2014; 2016). As there is no straightforward method of computing power in a full 2×5 within-subject ANOVA (Faul et al., 2007) we only made sure that sample size was adequate (i.e., sensitive) to detect small effects in a 1×5 within-subject design.

3.1.4 | Procedure

We included all 3464 clips (1513 with fans and 1951 without fans) and randomly sampled 5 clips for each of the 10 experimental categories (i.e., a total of 50 clips per participant). This procedure ensured that every participant was presented with a different, random series of 50 stimuli. Hence, the results of this experiment did not depend on a specific combination of the stimuli. The experiment was programmed as a desktop version with the PsychoPy2 software (Meinold & Furley, 2018). Every stimulus had to be rated on a score, dominance, pride and confidence dimension. Participants were asked to estimate the score and had to rate the soccer players in terms of how dominant, proud and self-confident the depicted player appeared. Therefore, a digital slider bar was presented with a 200 points scale, where the left pole represented -100 points (the lowest possible estimate) and the right pole represented 100 points (the highest possible estimate). The starting point was always in the middle of the bar (0) and the participants had to move the mouse cursor along the bar for their rating and log in the estimated score by clicking the left mouse button.

After providing informed consent, participants indicated their age, gender and soccer experience; they were also asked to indicate their average soccer consumption on television in hours per year. Subsequently, participants were instructed to observe the NVB of the player shown in each video clip and to estimate whether that player's team was leading or trailing at the time of the clip, as well as how

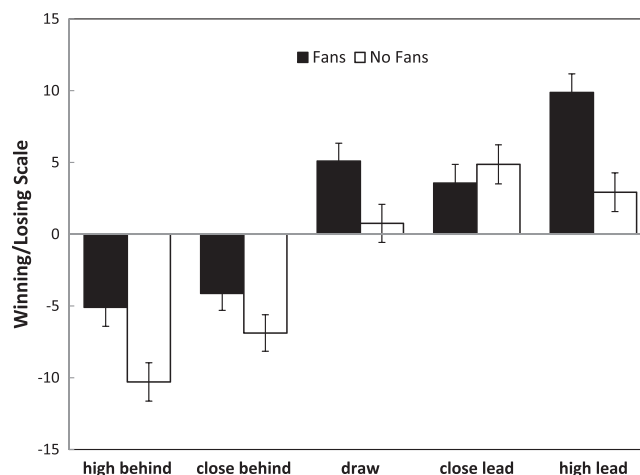


FIGURE 2 Mean ($-100 = \text{very low}$; $100 = \text{very high}$) ratings as a function of score category in Study 1. Error bars represent standard errors

dominant, how proud, and how confident he appeared. All videos were presented without sound since the noise would give a clear indication of whether a clip was taken from a normal or a ghost match.

3.2 | Results

The descriptive results of Study 1 are displayed in Figure 2 (<https://osf.io/xpcrz/> for data files). As all four dependent variables score, dominance, pride and confidence were highly correlated (Cronbach $\alpha = 0.86$) we decided to average the four dependent variables to represent one scale of winning/losing NVB (see online appendix for figures showing detailed descriptive results of all four variables).

A 2 (spectators vs. no spectators) $\times 5$ (behind by a lot, behind, tied score, ahead and ahead by a lot) ANOVA revealed a significant main effect of score category on the averaged winning/losing scale ($F[4, 960] = 52.301, p < .001, \eta^2 p = 0.179, 90\% \text{ CI } [0.142; 0.212]$). Regarding hypothesis 1, results confirm the linear relationship between participants' ratings and score categories across both fan conditions, indicating that ratings were more negative when the athletes were trailing than when they were leading (linear contrast: $F[1, 240] = 162.547, p < .001, \eta^2 p = 0.404, 90\% \text{ CI } [0.327; 0.470]$).

Regarding hypotheses 2 and 3, the results are not as straightforward. The fact that the main effect was present both in the spectator (linear contrast: $F[1, 240] = 94.784, p < .001, \eta^2 p = 0.283, 90\% \text{ CI } [0.206; 0.354]$) and in the no-spectator condition (linear contrast: $F[1, 240] = 98.702, p < .001, \eta^2 p = 0.291, 90\% \text{ CI } [0.214; 0.363]$) and shows similar effect sizes in the linear contrast analyses support hypothesis 2. However, there was a significant main effect for fans in the stadium on the averaged winning/losing scale ($F[1, 240] = 20.686, p < .001, \eta^2 p = 0.079, 90\% \text{ CI } [0.033; 0.138]$). This main effect shows that participants' ratings were slightly more positive in the fans condition as compared to the no-fans condition. In addition, the two-way interaction between the presence of fans and score category reached

² To control for team strength in the respective score categories we tested for the end-of-season ranking of the teams shown across the five experimental categories in a 1×5 ANOVA. This analysis showed no significant differences in team strength prevalent across the five categories.

significance for the averaged winning/losing scale ($F[4, 960] = 3.780$, $p = .005$, $\eta^2 p = 0.016$, 90% CI [0.003; 0.027]). To shed further light on this interaction, we ran a series of planned contrasts between the fans versus no fans in the respective score categories. Significant differences for the averaged winning/losing scale were evident in the high behind condition ($p < .001$; $d = 0.206$, 95% CI [0.078; 0.333]), draw ($p = .004$; $d = 0.172$, 95% CI [0.045; 0.299]), and high lead condition ($p < .001$; $d = 0.284$, 95% CI [0.155; 0.412]). No significant differences emerged in the close trailing ($p = .071$; $d = 0.117$, 95% CI [-0.010; 0.243]) and leading conditions ($p = .479$; $d = -0.046$, 95% CI [-0.172; 0.081]). Together with a visual analysis of Figure 2, this can be regarded as tentative support for hypothesis 3.

3.2.1 | Individual difference analyses

None of the indicators of soccer experience significantly moderated the results (when entered as a covariate). The categorical variables gender and soccer experience (active soccer experience or not) also did not significantly moderate the pattern of results when entered as a between-subject factor in the main analyses.

3.3 | Discussion

In Study 1 the four dependent measures (score guesses, ratings of dominance, pride and confidence) were highly correlated and therefore averaged to resemble a measure of winning/losing NVB. First, we were able to replicate the findings from Furley and Schweizer (2014, 2016) in the sport of soccer by demonstrating a significant main effect of score category across score guesses, dominance, pride and confidence ratings (support for hypothesis 1). This effect occurred in the spectator and no-spectator condition (support for hypothesis 2). In general, participants estimated soccer players playing in front of fans to be leading slightly higher, to be more dominant, prouder and more confident, no matter what the current score was. Therefore, the presence of fans in the stadium seems to have affected how dominant, proud and confident athletes appeared to observers. This finding was unexpected. Finally, there was a significant interaction between score category and the presence of fans. The significant pairwise comparisons in the high behind, draw and high lead categories explained this interaction and might be interpreted in line with the reasoning of Buck et al. (1992), who suggested that pleasant feelings (i.e., being in a high lead) coincided with social facilitation (i.e., greater expressiveness of 'positive NVB' in the context of competitive sports; in this case status-related NVB [cf. Furley & Schweizer, 2016]) and negative feelings (trailing) with social inhibiting (i.e., less expressiveness of 'negative NVB' in the context of competitive sports; in this case status-related NVB [cf. Furley & Schweizer, 2016]) in response to the presence of observers (tentative support for hypothesis 3). Nevertheless, this interpretation of the significant interaction does not perfectly fit the data, especially regarding the close behind and close lead category, and therefore leaves considerable room for speculation and further empirical research.

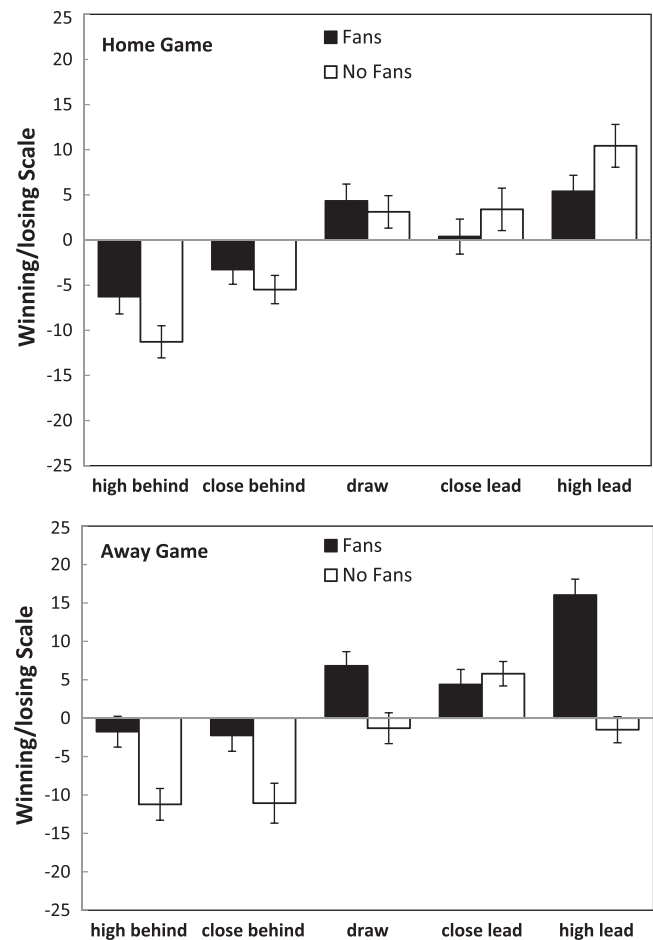


FIGURE 3 Study 1 Winning/Losing scale further distinguished by home and away games. Error bars represent standard errors

Although the pattern of results clearly shows differences in how observers interpret the NVB of athletes depending on the score line and the presence of fans, there might be a further variable of interest that is obscured in the present analyses: that is, the game location, as this will influence what type of people are mainly watching the game in the spectator condition and how they may behave during the games. As the game location was known for every experimental stimulus, we visually explored potential differences as a consequence of game location in Figure 3. These additional analyses provided first indications that the interaction between presence of fans and score category depended on whether the fans were a supporting crowd (as in home games) or a non-supporting crowd (as in away games). Figure 3 shows these differences visually by further distinguishing home games (top column) and away games (below column). Although these descriptive differences seem interesting, caution is warranted on drawing inferences from these data as the random allocation procedure of the software only ensured even allocation to the experimental score and spectator categories. The distribution was not even for home and away games and, hence it was problematic to analyse data with inferential statistical tests. Therefore, we decided to corroborate the findings depicted in Figure 3

in an additional experiment and refrain from explicitly discussing the findings presented in Figure 3.

4 | STUDY 2: SCORE ESTIMATION AND EVALUATIVE CODING DEPENDING ON SCORE LINE, PRESENCE OF FANS AND GAME LOCATION

4.1 | Method

4.1.1 | Participants

A total of 160 participants (90 male, 69 female, 1 diverse; $M_{age} = 31.26$ years; $SD = 13.86$ years) participated in Study 2. Participants gave written informed consent before the experiment. Fifty four participants had no active soccer experience, 57 reported to have some recreational soccer experience and 49 played on a team (1 of them professionally). The participants consumed an average of 41.80 hours ($SD = 62.91$) of soccer per year on television. Participants were recruited on a voluntary basis by research assistants. Again, power considerations (Faul et al., 2007) only made sure that sample size was adequate (i.e., sensitive) to detect small effects in a 1×5 within-subject design.

4.1.2 | Procedure

Everything was identical to Study 1 except that this time all 3464 clips (1513 with fans and 1951 without fans) were further divided depending on whether they were showing the home team or the away team. In addition, we excluded the close lead and close behind category to simplify the experimental design and avoid presenting participants with too many videos. Hence, we had a 2 (fans vs. no fans) $\times$ 2 (home vs. away) $\times$ 3 (high behind vs. draw vs. high lead) experimental design. We randomly sampled 4 clips for each of the 12 experimental categories (i.e., a total of 48 clips per participant). This procedure ensured that every participant was presented with a different, random series of 48 stimuli. Hence, the results of this experiment did not depend on a specific combination of the stimuli. The experiment was programmed as a desktop version with the PsychoPy2 software (Meinold & Furley, 2018) and used the same score, dominance, pride and confidence scales as Study 1.

4.2 | Results

The descriptive results of Study 2 are displayed in Figures 4 and 5 (<https://osf.io/xpcrz/> for data files). As again all four dependent variables score, dominance, pride and confidence were highly correlated (Cronbach $\alpha = 0.85$) we decided to average the four dependent variables to represent one scale of winning/losing NVB (see online appendix for figures showing detailed descriptive results of all four variables).

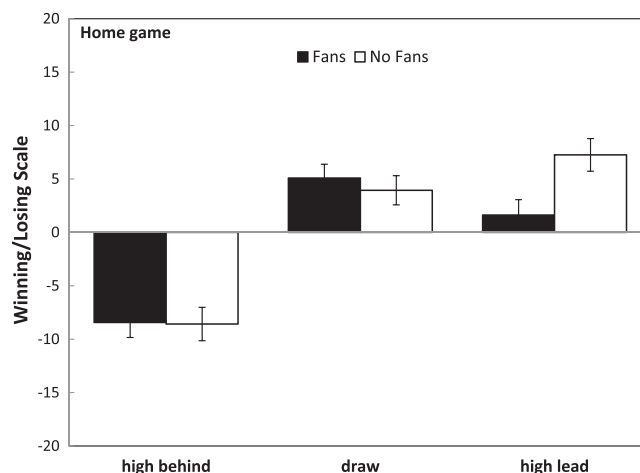


FIGURE 4 Study 2 ratings for home games as a function of score and spectator presence. Error bars represent standard errors

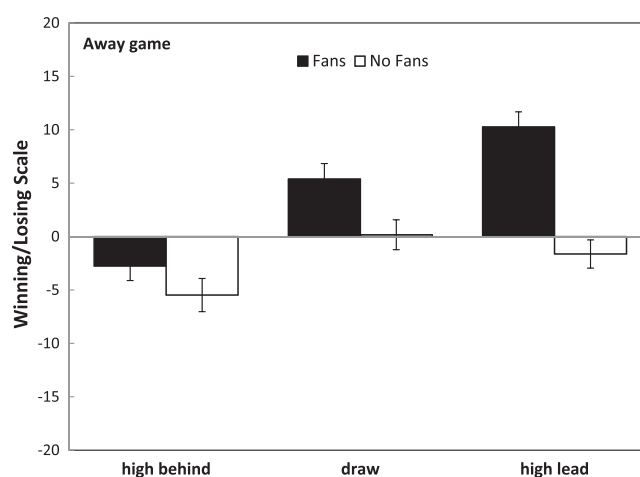


FIGURE 5 Study 2 ratings for away games as a function of score and spectator presence. Error bars represent standard errors

A 2 (spectators vs. no spectators) $\times$ 2 (home vs. away) $\times$ 3 (behind by a lot, tied score, and ahead by a lot) ANOVA revealed a significant main effect (support for hypothesis 1) of score category on the winning/losing scale ($F[2, 318] = 73.893, p < .001, \eta^2 p = 0.317, 90\% \text{ CI } [0.248; 0.377]$; linear contrast: $F[1, 159] = 107.300, p < .001, \eta^2 p = 0.403, 90\% \text{ CI } [0.307; 0.482]$). Further, there was a significant main effect for fans in the stadium on the winning/losing scale ($F[1, 159] = 11.700, p < .001, \eta^2 p = 0.069, 90\% \text{ CI } [0.018; 0.139]$). There was no significant main effect for location of game on the winning/losing scale ($F[1, 159] = 1.256, p = .264, \eta^2 p = 0.008, 90\% \text{ CI } [0.000; 0.045]$).

The two-way interaction between the presence of fans and game location was significant ($F[2, 318] = 30.322, p < .001, \eta^2 p = 0.160, 90\% \text{ CI } [0.100; 0.217]$). The two-way interaction between score category and game location was also significant ($F[2, 318] = 5.489, p = .005, \eta^2 p = 0.033, 90\% \text{ CI } [0.006; 0.068]$). The two-way interaction between the presence of fans and score category did not reach significance ($F[2, 318] = 0.564, p = .569, \eta^2 p = 0.004, 90\% \text{ CI } [0.000; 0.017]$). Finally, the

three-way interaction between the presence of spectators and game location and score category reached significance ($F[2, 318] = 8.482$, $p < .001$, $\eta^2 p = 0.051$, 90% CI [0.016; 0.091]).³

To shed more light on the significant interactions and explore hypothesis 4, we ran additional 2-way ANOVAs.

4.2.1 | Differences between home and away

Home games

A 2 (spectators vs. no spectators) $\times$ 3 (behind by a lot, tied score ahead by a lot) ANOVA revealed a significant main effect of score category on the averaged winning/losing scale ($F[2, 318] = 64.131$, $p < .001$, $\eta^2 p = 0.287$, 90% CI [0.218; 0.348]). This result confirms the linear relationship between participants' ratings and score categories for the home games across both fan conditions, indicating that ratings were more negative when the athletes were trailing than when they were leading (linear contrast: $F[1, 159] = 79.884$, $p < .001$, $\eta^2 p = 0.334$, 90% CI [0.238; 0.419]). There was no main effect for fans in the stadium during home games on the averaged winning/losing scale ($F[1, 159] = 1.780$, $p = .184$, $\eta^2 p = 0.011$, 90% CI [0.000; 0.052]). The two-way interaction between the presence of fans and score category during home games reached significance for the averaged winning/losing scale ($F[2, 318] = 3.530$, $p = .030$, $\eta^2 p = 0.022$, 90% CI [0.001; 0.051]).

To shed further light on this interaction, we ran a series of planned contrasts between the fans versus no fans in the respective score categories (see Figure 4). Significant differences for the averaged winning/losing scale were only evident in the high lead condition ($p = .003$; $d = -0.234$, 95% CI [-0.391; -0.077]) between matches with spectators compared to without spectators for home games. No significant differences emerged in the draw ($p = .526$; $d = 0.050$, 95% CI [-0.105; 0.205]) and high trailing condition ($p = .944$; $d = 0.006$, 95% CI [-0.149; 0.161]).

Away games. A 2 (spectators vs. no spectators) $\times$ 3 (behind by a lot, tied score, ahead by a lot) ANOVA revealed a significant main effect of score category on the averaged winning/losing scale ($F[2, 318] = 20.043$, $p < .001$, $\eta^2 p = 0.112$, 90% CI [0.060; 0.165]) for away games. This result confirms the linear relationship between participants' ratings and score categories for the away games across both fan conditions, showing that ratings were more negative when the athletes were trailing than when they were leading (linear contrast: $F[1, 159] = 34.856$, $p < .001$, $\eta^2 p = 0.180$, 90% CI [0.097; 0.265]). This time there was a significant main effect for fans in the stadium during away games on the averaged winning/losing scale ($F[1, 159] = 42.274$, $p = .001$, $\eta^2 p = 0.210$, 90% CI [0.123; 0.297]). The two-way interaction

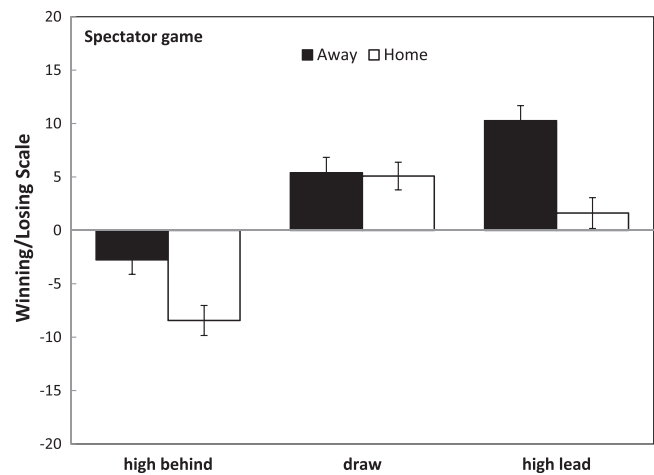


FIGURE 6 Study 2 ratings for spectator games as a function of score and game location. Error bars represent standard errors

between the presence of fans and score category during away games also reached significance for the averaged winning/losing scale ($F[2, 318] = 6.042$, $p = .003$, $\eta^2 p = 0.037$, 90% CI [0.008; 0.073]).

To shed further light on this interaction, we ran a series of planned contrasts between the fans versus no fans in the respective score categories (see Figure 5). Significant differences for the averaged winning/losing scale were only evident in the high lead ($p < .001$; $d = 0.513$, 95% CI [0.347; 0.677]) and draw condition ($p = .003$; $d = 0.235$, 95% CI [0.077; 0.391]) between matches with spectators compared to without spectators for away games. No significant differences emerged in the high trailing condition ($p = .181$; $d = 0.106$, 95% CI [-0.049; 0.261]).

4.2.2 | Differences between spectator and ghost games

Spectator games

A 2 (home vs. away) $\times$ 3 (behind by a lot, tied score, ahead by a lot) ANOVA revealed a significant main effect of score category on the averaged winning/losing scale ($F[2, 318] = 45.287$, $p < .001$, $\eta^2 p = 0.222$, 90% CI [0.156; 0.282]; linear contrast: $F[1, 159] = 61.559$, $p < .001$, $\eta^2 p = 0.279$, 90% CI [0.185; 0.366]). There was also a main effect for game location for the spectator games on the averaged winning/losing scale ($F[1, 159] = 24.046$, $p < .001$, $\eta^2 p = 0.131$, 90% CI [0.059; 0.213]), showing more positive ratings for away games. The two-way interaction between the game location and score category during spectator games also reached significance for the averaged winning/losing scale ($F[2, 318] = 5.782$, $p = .003$, $\eta^2 p = 0.035$, 90% CI [0.007; 0.071]).

To shed further light on this interaction, we ran a series of planned contrasts between the away versus home conditions in the respective score categories (see Figure 6). Significant differences for the averaged winning/losing scale were evident in the high trailing ($p < .001$; $d = 0.288$, 95% CI [0.130; 0.446]) and high lead condition ($p < .001$; $d = 0.391$, 95% CI [0.230; 0.551]) between matches with

³ Similar to Study 1, none of the indicators of soccer experience significantly moderated the results (when entered as a covariate) in Study 2. The categorical variables gender and soccer experience (active soccer experience or not) also did not significantly moderate the pattern of results when entered as a between-subject factor in the main analyses.

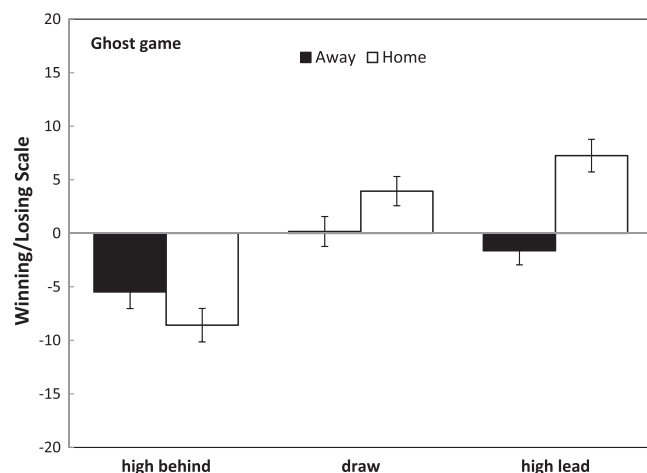


FIGURE 7 Study 2 ratings for ghost games as a function of score and game location. Error bars represent standard errors

spectators compared to without spectators for home games. No significant differences emerged in the draw condition ($p = .873$; $d = 0.013$, 95% CI $[-0.142; 0.168]$).

Ghost games. A 2 (home vs. away) $\times$ 3 (behind by a lot, tied score, ahead by a lot) ANOVA revealed a significant main effect of score category on the averaged winning/losing scale ($F[2, 318] = 32.534$, $p < .001$, $\eta^2 p = 0.170$, 90% CI $[0.109; 0.228]$; linear contrast: $F[1, 159] = 51.483$, $p < .001$, $\eta^2 p = 0.245$, 90% CI $[0.153; 0.332]$). There was also a main effect for game location for the spectator games on the averaged winning/losing scale ($F[1, 159] = 8.247$, $p = .005$, $\eta^2 p = 0.049$, 90% CI $[0.009; 0.113]$), but this time showing slightly more positive ratings for home games. The two-way interaction between the game location and score category during spectator games also reached significance for the averaged winning/losing scale ($F[2, 318] = 7.906$, $p < .001$, $\eta^2 p = 0.047$, 90% CI $[0.014; 0.087]$).

To shed further light on this interaction, we ran a series of planned contrasts between the away versus home conditions in the respective score categories (see Figure 7). Significant differences for the averaged winning/losing scale were evident in the draw ($p = .040$; $d = -0.164$, 95% CI $[-0.320; -0.008]$) and high lead condition ($p < .001$; $d = -0.351$, 95% CI $[-0.510; -0.190]$) between matches with spectators compared to matches without spectators for home games. No significant differences emerged in the high trailing condition ($p = .188$; $d = 0.104$, 95% CI $[-0.051; 0.260]$).

4.3 | Discussion

Study 2 also demonstrated a significant main effect of score category on the averaged winning/losing scale (support for hypothesis 1). While the main effects and interactions for the spectator and location factors are not trivial to interpret, and we do not have an explanation for all significant effects, we will focus on what we consider the most striking effects and start with the significant three-way interaction. This three-

way interaction is mainly explained by diverging patterns of ratings as a function of both fans present and game location in the high behind (negative valence) and high lead (positive valence) categories (see Figures 4 and 5). However, before we discuss this pattern, we do want to point out that soccer players were also rated more positively when the score was tied at away games with (non-supportive) fans present in comparison to ghost games. This resembles the pattern when they were in a high lead at away games with a (non-supportive) crowd present in comparison to ghost games. This might suggest that both leading and drawing away is experienced as a positive achievement and leads to a higher non-verbal expressiveness in front of a non-supportive crowd, or, stated differently, soccer players might get a non-verbal boost from a non-supportive crowd when they are playing well/beating the home team. On the other hand, playing at home did not have this effect in the high lead condition; in this case the presence of a supportive audience seems to have led to less non-verbal expressiveness (at least this is how it was rated by observers) in comparison to when no fans were present. Potentially this could be interpreted as displaying more modest NVB when achieving something positive in front of a supportive audience in comparison to no audience present.

A further finding, which was evident in Study 2, was that trailing by far was rated more negatively at home than away, particularly if fans were present in the stadium. This is probably caused by the evident home advantage effect in sports, which suggests that it is more likely to be successful at home in comparison to away. Hence, trailing by far might be experienced as more negative at home compared to away leading to more negative NVB which is rated as less dominant, less proud and less confident, particularly if fans are present (as there were no significant differences in the no-fans condition). Taken together, the pattern of results does not support hypothesis 4 (i.e., greater expressiveness of both positive and negative feelings in home games as compared to away games), but shows complex changes of athletes NVB in response to a combination of situational variables (score line, spectator presence and game location).

5 | GENERAL DISCUSSION

The goal of the present investigation was to utilize the unique conditions of the Covid-19 pandemic to investigate the effects of the absence of spectators in sports competitions on the NVB of soccer players in a natural experiment and thereby to contribute to the important question in psychology of 'how the presence of others affects people's behaviour'. In two studies, we were able to replicate and extend the findings from Furley and Schweizer (2014, 2016) in the sport of soccer by demonstrating a linear relationship between participants' ratings and score categories, indicating that ratings were more negative when the athletes were trailing (less dominant, less proud and less confident) than when they were leading. Further, the results of Study 1 provided unique support for social psychological theorizing that suggests that pleasant feelings (i.e., being in a high lead) tend to be expressed non-verbally to a greater extent in the presence of others (social facilitation) and negative feelings (trailing) tend to be expressed

less in the presence of observers (social inhibition). However, in line with previous research suggesting that the relationships of observing people have to be considered when investigating audience effects on NVB (e.g., Wagner & Lee, 1999), Study 2 provided unique support, showing that a supporting crowd during home games can have a different influence on the NVB of soccer players than a non-supportive away crowd. The most striking finding from Study 2 was that leading at away games led to higher non-verbal expressiveness in the presence of a non-supportive crowd, whereas leading at home games seems to have led to less non-verbal expressiveness in the presence of a supportive audience (see Figures 4 and 5).

The findings presented are in line with previous research on NVB in sports (Furley, 2019; Furley & Schweizer, 2020) and the thin slices of behaviour hypothesis (Ambady et al., 2000; Ambady & Rosenthal, 1993), and add significantly to the literature on audience effects on non-verbal expressiveness (Wagner & Lee, 1999 for a review) and NVB in sports (Furley, 2021). The present data might be interpreted as in line with the behavioural ecology theory of NVB (Crivelli & Fridlund, 2018; Fridlund, 1994) as the NVB of soccer players was influenced by a range of contextual factors (current score, presence of an audience and game location) and interpreted accordingly by observers in a thin-slices experiment. The present research showed that leading and trailing in soccer is associated with non-verbal cues that are interpreted as signs of winning and losing. Thin slices of behaviour (very short recordings of athletes' behaviour) were sufficient to permit quite precise estimates of the actual score in sport competitions. Our results suggest that leading soccer players change their NVB compared to trailing athletes along the verticality dimension of social relations (Burgoon & Hoobler, 2002; Hall & Friedman, 1999; Hall et al., 2005) by showing more dominance, more pride and more confidence. In this respect, we were able to replicate (hypothesis 1) the results of Furley & Schweizer (2014, 2016). We consider replications like this one a research necessity and they should not be regarded a trivial research step considering the increasing calls for replication in the psychological literature (Open Science Collaboration, 2015; Pashler & Wagenmakers, 2012; Simons, 2014; Yong, 2012).

However, the present studies also extend previous research. First, they show that the reasoning of Furley & Schweizer (2014) applies to a further context (i.e., soccer) and, importantly, they show that leading and trailing in soccer during ghost games is also accompanied by non-verbal cues that are interpreted as signs of winning and losing by observers. This might be interpreted in line with the evolutionary reasoning of winning and losing NVB in sports (support of hypothesis 2), which has been argued to resemble the adaptive signalling of primates when winning or losing status contests (Furley, 2019; Furley & Schweizer, 2016). According to evolutionary accounts of non-verbal expressions, submissive non-verbal expressions originally served the function of avoiding further attacks in a fight. On the other hand, winning a fight leads primates to show dominant and proud NVBs, which are also believed to be adaptive, as they communicate their superiority and higher status over the opponent and thereby can also save valuable resources by preventing further aggressive encounters. Arguably, the fact that we found that athletes were rated

as more dominant and prouder when winning and more submissive and ashamed when losing also during ghost games and not only when large crowds of fans were present might be regarded as indicative of this evolutionary account of NVB in sports. However, we acknowledge that this is speculative and needs to be corroborated by further research. Also, there were important differences in the NVB of athletes as a function of the presence or absence of fans.

Perhaps the most important finding from the present series of studies was the support for the theorizing of Buck et al. (1992; see also Wagner & Lee, 1999 for a review) in an ecologically valid context (support for hypothesis 3). Buck and colleagues (1992) suggest that a distinguishing feature between studies showing social facilitation or social inhibiting of non-verbal expressions might be the affective nature of the situations with pleasant feelings coinciding with social facilitation and negative feelings with social inhibiting. This pattern was clearly evident in Study 1 as pleasant feelings (i.e., being in a high lead) of soccer players led to higher observer ratings on the inferred score, and on perceived confidence, pride and dominance when the player was playing in front of spectators in comparison to an empty stadium. Arguably this might be interpreted to mean that soccer fans in the stadium led to social facilitation effects on players who were feeling well because they were in the lead and therefore expressed more status-related NVB (i.e., more dominance, pride, confidence). On the other hand, expressions of negative feelings when trailing by far seem to have been suppressed (i.e., social inhibition) when spectators were present in comparison to when no spectators were present as observer ratings were comparably higher (i.e., less negative). However, it is important to note that it is not clear how comparable the naturalistic experiment settings of the present research are to prior studies on social facilitation and social inhibition of emotional expressions as the soccer players in the present study are still playing with and against other players and other officials and staff are also watching during the 'ghost games'. Moreover, the matches are video recorded and broadcasted to large audiences watching the games on television, of which the players are obviously aware. Hence, there remains plenty of social attention towards the players also when they are not playing in front of large audiences (see Steinmetz & Pfattheicher, 2017). A main distinguishing factor between games with spectators and games without spectators seems to be that there is not as much direct social feedback from fans in terms of cheering or booing in response to an athlete's behaviour during 'ghost games'.

In addition, several lines of investigation have found evidence for interaction effects between the relationship to the observer and the type of emotion in predicting social facilitation or inhibition of NVB (Buck et al., 1992; Fridlund, 1991; Fridlund et al., 1990). This general theorizing was also confirmed in Study 2. Although the three-way interaction between score line (as a proxy for valence) $\times$ presence of fans (as a proxy for social audience) $\times$ game location (as a proxy of supportive or non-supportive audience) is complicated to disentangle and does not support hypothesis 4 (see Figures 4–7), several intriguing findings emerged. The three-way interaction in Study 2 seems to be mainly explained by diverging patterns of ratings as a function of both fans present and game location in the high lead (positive valence) categories

(see Figures 4–7). Leading at away games led to a higher expressiveness of ‘positive NVB’ in the presence of a non-supportive crowd, whereas leading at home games seems to have led to less expressiveness of ‘positive NVB’ in the presence of a supportive audience. While there is plenty of room for speculation about this surprising finding, one way of explaining this pattern might be with reference to findings on the home advantage in sports (e.g., Furley et al., 2018). Research on the home advantage in general shows that athletes are more successful at home games than away games. Therefore, leading at home is more likely than leading at away games in soccer (especially being in a high lead), and arguably represents a ‘more normal’ achievement, whereas being in a high lead away would represent a ‘more special’ achievement. Hence, home-playing soccer players might somewhat inhibit their ‘positive NVB’ when being in a high lead in front of a supportive audience as this is not ‘such a great achievement’ and they might want to appear somewhat modest (‘not bragging’). Whereas the opposite seems to occur at away games, in which away-playing soccer players show social facilitation of ‘positive NVB’ in front of a non-supportive audience. Perhaps these away-playing athletes are triggered by the non-supportive crowd to show a greater expressiveness of their ‘positive NVB’ in the case of the less likely scenario of beating their opponent at an away game. Nevertheless, we acknowledge that further confirmatory research is warranted to gain confidence in this speculative post-hoc explanation.

The present research has notable strengths and weaknesses. One strength of the present research is that the studies were high-powered within-subject designs that were sensitive enough to reliably detect small effects. Furthermore, not all observers were shown the same stimuli, but different participants were randomly assigned to different subsets of all stimuli that were randomly drawn from a large sample of over 3000 stimuli. This approach is likely to reduce the probability that effects are dependent on one particular set of a highly selected set of stimuli, as called for in psychological research (Fiedler, 2011). A further strength of the present approach is its high external validity. As the stimuli and setting are not artificially created—which is typical for the majority of research on audience effects of NVB (Buck et al., 1992; Fridlund, 1991; Fridlund et al., 1990)—but recordings of real sports competitions, the results are likely to transfer to field settings and might arguably be even be stronger in the field when watching real live NVB in a soccer stadium given the influence of crowd dynamics like emotional contagion (Furley, 2019). Therefore, we consider it likely that the present findings might be relevant in further everyday settings, but acknowledge that this needs to be tested by additional research.

A limitation of our research was that athletes were not evenly distributed across experimental stimulus categories similar to the previous thin-slices studies in sport (Furley & Memmert, 2021; Furley & Schweizer, 2014; 2016). Hence, we cannot entirely rule out that certain stimuli or constellations of stimuli account for, or contribute to, the pattern of results. We tried to partially address this issue by our random selection procedure from our stimulus battery. Another limitation of the natural experimental design are potential confounding variables with our experimental categories. It is possible that further variables might have led to differences in the evaluations of athlete’s

NVB that were not accounted for in the study designs, particularly in the spectator vs. no-spectator condition. Although we consider this unlikely it is possible that it was not so much the presence or absence of fans that was correlated to differential effects on the evaluation of athletes’ NVB, but for example that athletes were affected by other variables that came alongside political Covid measures, like being more isolated due to the lockdown or worried about the progression of the pandemic. However, as the data did not simply show that athlete evaluations were always more negative when no spectators were present, but differed depending on the specific context, we consider such alternative interpretations of the data unlikely.

In conclusion, the present research highlights that athletes change their NVB depending on a range of contextual (game location, presence of spectators) and situational factors (score line), and observers are well equipped to pick up subtle changes in NVB that occur during sport competitions. In this respect, it seems likely that the NVB of athletes contributes to explaining performance effects associated with the home advantage and social facilitation/inhibition as observers like opponents, team-mates, referees and spectators might be influenced by differential non-verbal demeanours.

AUTHOR CONTRIBUTIONS

Philip Furley, Nina Riedl and Babett Lobinger developed the study concept. Data collection was performed by Philip Furley and Nina Riedl. The data analysis was performed by Philip Furley and Nina Riedl. All authors contributed to the interpretation of the results. Philip Furley drafted the manuscript, Nina Riedl and Babett Lobinger provided critical revisions. All authors approved the final version of the manuscript for submission.

ACKNOWLEDGMENTS

The research was not supported by any funding.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

All data can be found here: <https://osf.io/xpcrz/>

ETHICS STATEMENT

The study was approved by the local university’s ethics committee.

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Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Furley, P., Riedl, N., & Lobinger, B. (2023). Non-verbal behaviour of professional soccer players performing in the absence or presence of fans. *European Journal of Social Psychology*, 53, 1144–1156. <https://doi.org/10.1002/ejsp.2964>